Converting chemical expressions into word and chemical equations:

Convert the following chemical expressions to work and symbol equations. For each of the following, identify the reactants and products.

1. Hydrogen reacts explosively with oxygen to form water vapour liberating a lot of heat energy.

   Reactants: Hydrogen and oxygen
   Product: Water vapour and heat

   Word equation: hydrogen + oxygen → water vapour and heat
   Chemical equation: $H_2 + O_2 \rightarrow H_2O$ + heat

2. Sodium metal dissolves violently in water yielding sodium hydroxide (NaOH) and hydrogen gas.
3. Hydrochloric acid (HCl) combines with sodium hydroxide (NaOH) to produce sodium chloride (NaCl) and water.
4. Hydrogen gas combines with fluorine gas (F_2) to form hydrogen fluoride (HF).
5. The reaction of potassium chloride (KCl) and silver nitrate ($AgNO_3$) yields potassium nitrate (KNO_3) and silver chloride (AgCl)
6. The decomposition of sodium bicarbonate ($NaHCO_3$) produces sodium carbonate (Na_2CO_3), carbon dioxide, and water.
7. The reaction of aluminium sulphate ($Al_2(SO_4)_3$) and barium chloride ($BaCl_2$) produces aluminium chloride ($AlCl_3$) and barium sulphate ($BaSO_4$).
8. The combustion of ethane (C_2H_8) in oxygen gas yields carbon dioxide and water vapour.
9. Dinitrogen pentoxide (N_2O_5) gas decomposes to form nitrogen gas (N_2) and oxygen gas.
10. Cobalt sulphate ($CoSO_4$) reacts with lead nitrate ($Pb(NO_3)_2$) to form cobalt nitrate ($Co(NO_3)_2$) and lead sulfate ($Pb(SO_4)_3$).
11. Phosphoric acid (H_3PO_4) reacts with magnesium hydroxide ($Mg(OH)_2$) to yield magnesium phosphate ($Mg_3(PO_4)_2$) and water.
12. The reaction of iron bromide ($FeBr_3$) and ammonium sulphide ($(NH_4)_2S$) produces ammonium bromide (NH_4Br) and iron sulphide (Fe_2S_3)